

Ply Orientation Effects on Mode I Delamination in Laminated Composites: A Comparative Numerical Study of VCCT and CZM

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Abstract

Delamination is one of the most prevalent failure mechanisms in laminated fiber-reinforced composites, leading to substantial degradation in stiffness and load-carrying capacity. Accurate numerical prediction of delamination initiation and growth is therefore essential for the reliable design of composite structures. This study presents a comparative numerical investigation of two widely used fracture modeling approaches—the Virtual Crack Closure Technique (VCCT) and the Cohesive Zone Method (CZM)—for Mode I delamination in carbon-fiber-reinforced polymer (CFRP) laminates with varying unidirectional ply orientations. Rather than focusing on methodological developments, the study emphasizes practical insights into the relative strengths and limitations of each approach when applied to laminated composites. Finite element simulations of double cantilever beam (DCB) specimens are conducted and verified through analytical solutions and systematic mesh convergence studies. The influence of ply orientation on delamination behavior is examined using load-displacement responses, crack propagation characteristics, and global energy evolution. The results indicate that laminate orientation has a pronounced effect on fracture response, with 0° plies sustaining higher load levels, while 90° plies exhibit larger crack opening displacements. Although VCCT offers computational efficiency and reduced mesh sensitivity, it exhibits limitations in maintaining consistent energy evolution during progressive crack growth, as evidenced by a systematic discrepancy between internal energy and applied external work. In contrast, CZM formulations provide a more stable and physically consistent representation of energy dissipation throughout delamination. The comparative findings offer practical guidance for selecting appropriate fracture modeling strategies for delamination analysis in laminated composite structures.

Keywords: Laminated composites; Ply orientation; Mode I delamination; Virtual Crack Closure Technique; Cohesive Zone Method; Finite element analysis

Highlights

- Comparative numerical assessment of VCCT and CZM for Mode I delamination in CFRP laminates.
- Finite element models are verified using analytical solutions and systematic mesh convergence studies.
- The investigated methods exhibit similar crack growth and crack opening displacement trends.
- VCCT offers higher computational efficiency, while CZM provides improved consistency in energy evolution.
- Ply orientation is shown to significantly influence delamination initiation and propagation behavior.

1 Introduction

Laminated fiber-reinforced composites enable high specific stiffness and strength together with tailorable anisotropy, making them attractive for weight-critical structural systems [6]. At the microscale, such composites consist primarily of reinforcing fibers embedded within a continuous polymer matrix with a thin interphase region governing load transfer between the fibers and the surrounding matrix [13, 37]. In laminated composites, adjacent plies are bonded through resin-rich interlaminar regions rather than discrete adhesive layers, and these interfaces often represent the weakest links under out-of-plane loading. These attributes have led to their widespread adoption in the aeronautical and automobile sectors, where structural mass optimization and mechanical reliability are crucial. In recent years, their application has expanded to wind turbine blades [59], biomedical implants [84], and unmanned aerial vehicles [28]. However, the layered architecture of laminated composites can promote interlaminar damage in the form of delamination, defined as separation between adjacent plies driven by weak interfacial strength and out-of-plane loading [5, 12, 30, 51, 75]. Once initiated, delamination can propagate with limited external indication and can cause substantial loss of stiffness and residual strength, motivating extensive experimental and numerical research on interlaminar fracture [45, 77].

Mode I delamination in the double cantilever beam (DCB) configuration is a canonical benchmark because it provides a controlled crack-driving mechanism and supports analytical solutions for validation [81]. Although standardized experiments (e.g., DCB, end-notched flexure (ENF), and mixed-mode bending (MMB)) provide fracture toughness and crack-growth data [35, 65, 69, 77], experimental programs are often costly, time-consuming, and are not well suited for systematic studies of numerical sensitivity, such as mesh refinement and formulation-dependent differences. Consequently, finite element (FE) modeling remains essential for predictive modeling, parametric studies, and interpretation of fracture experiments [27, 75].

Among computational approaches for delamination, two families dominate computational mechanics practice: (i) energy-based methods such as the Virtual Crack Closure Technique (VCCT) and (ii) traction-based methods such as cohesive zone modeling (CZM). VCCT evaluates the strain energy release rate from nodal forces and displacement jumps near the

crack tip, consistent with linear elastic fracture mechanics [52, 70]. In contrast, CZM represents fracture through a traction–separation law with damage initiation and evolution over a finite fracture process zone, enabling progressive stiffness degradation and crack propagation without a predefined crack-growth criterion [24, 31, 36, 64, 79, 83]. Within commercial FE implementations, CZM can be realized using discrete cohesive elements (CE) or surface-based cohesive interactions (cohesive surfaces, CS). While these approaches are widely available, their numerical behavior differs markedly. VCCT typically exhibits weak mesh sensitivity once the crack-tip region is adequately resolved, whereas cohesive formulations can be strongly influenced by interface discretization, penalty stiffness, and the ability of the mesh to resolve the cohesive process zone length [38, 56, 57, 61, 79].

Despite extensive literature on VCCT and CZM, direct quantitative comparison among VCCT, CE, and CS remains difficult because many published studies differ in geometry, material systems, cohesive parameter selection, mesh refinement protocols, and solver controls. Comparative investigations exist for selected configurations and subsets of methods [32, 33, 39, 50, 58, 66], yet there remains a practical gap. The literature provides limited unified formulation-consistent guidance on (i) the mesh resolution required to obtain mesh-independent peak loads, (ii) agreement with analytical Mode I benchmarks under controlled assumptions, and (iii) accuracy–cost trade-offs that determine the feasibility of large-scale simulations and parametric sweeps [9, 16]. In addition, many studies emphasize load–displacement agreement alone, while providing limited discussion of global energy consistency (external work versus internal energy) and the computational implications of contact-based surface formulations [19, 20, 29]. As a result, it remains difficult to distinguish formulation-driven differences from numerical artifacts introduced by inconsistent discretization and solver choices.

This work addresses these needs by establishing a controlled benchmark comparison of three commonly used delamination modeling strategies in Abaqus: VCCT, CE, and CS, applied to Mode I delamination in a DCB configuration. All simulations are performed using identical geometry, material properties, fracture toughness, cohesive parameters, mesh compatibility rules, and solver tolerances to isolate formulation-driven effects. Mesh convergence is quantified using the peak (critical) load as the primary metric for delamination initiation, and converged predictions are validated against a closed-form analytical Mode I solution [86]. Beyond peak load validation, the study compares (i) the full load–displacement response, (ii) global energy measures obtained from Abaqus history outputs to interpret differences between energy-based and traction-based fracture representations, (iii) crack length evolution as a function of crack mouth opening displacement (CMOD) [11, 85], (iv) computational cost under identical hardware and solver settings, and (v) the influence of interface ply orientation from $0^\circ/0^\circ$ to $90^\circ/90^\circ$ across all formulations [3, 4, 8, 67, 73]. The contribution of this paper is not the introduction of a new fracture model, but the establishment of a rigorously controlled, formulation-consistent benchmark that quantifies mesh requirements and accuracy–efficiency trade-offs for widely used delamination approaches. The resulting guidance supports more defensible selection of delamination modeling strategies for both research-grade simulations and engineering analyses, and provides a validated baseline for future extensions involving mixed-mode loading and more complex interfacial constitutive behavior [35].

2 Methodology

2.1 Theoretical Background for Mode I Delamination

This study investigates Mode I delamination using established fracture mechanics and cohesive interface formulations, implemented using the commercial finite element solver ABAQUS/Standard (Dassault Systèmes, 3DEXPERIENCE R2017x) [22]. The program offers several interface modeling options, including VCCT, CS, and CE formulations. The mathematical models and fracture criteria applied in the analysis are described in this section.

2.1.1 VCCT

The VCCT is a linear elastic fracture mechanics (LEFM) based approach implemented within commercial finite element solvers to evaluate the energy available for interlaminar crack growth in composite laminates [6, 10, 68]. VCCT assumes that a crack propagates when the strain-energy release rate G reaches the material fracture toughness G_c [14, 46–49]:

$$G \geq G_c. \quad (1)$$

A detailed derivation of the VCCT formulation from its continuum integral form to its discrete nodal representation is provided in Appendix A.

In the finite element setting, the method relates the nodal reaction force vector $\mathbf{F}^{cd}$ acting at the first node pair ahead of the crack tip to the relative opening displacement $\Delta \mathbf{u}^{cd} = \mathbf{u}^c - \mathbf{u}^d$ between the paired nodes behind the crack tip, as illustrated in Figure 1. Following Irwin’s classical formulation [44] and the numerical implementation described by Krueger [54], the virtual work associated with a small crack extension Δa is

$$W = \frac{1}{2} \mathbf{F}^{cd} \cdot (\mathbf{u}^c - \mathbf{u}^d), \quad (2)$$

where $\mathbf{F}^{cd} = (F_x^{cd}, F_y^{cd}, F_z^{cd})$ is the nodal reaction force vector at the node pair (c, d) ahead of the crack tip, and $\mathbf{u}^c$ and $\mathbf{u}^d$ are the displacements at the paired upper and lower crack-tip nodes behind the tip. The vertical component F_y^{cd} corresponds to the opening direction in Figure 1(a). The strain-energy release rate is then

$$G = \lim_{\Delta a \rightarrow 0} \frac{W}{\Delta A} = \lim_{\Delta a \rightarrow 0} \frac{1}{2b\Delta a} \mathbf{F}^{cd} \cdot (\mathbf{u}^c - \mathbf{u}^d), \quad (3)$$

where $\Delta A = b\Delta a$ is the newly created crack area and b is the specimen width.

Decomposing the force and displacement vectors into orthogonal components leads to the mode-specific energy release rates:

$$G_I = \lim_{\Delta a \rightarrow 0} \frac{1}{2b\Delta a} F_y^{cd} (u_2^c - u_2^d), \quad (4)$$

$$G_{II} = \lim_{\Delta a \rightarrow 0} \frac{1}{2b\Delta a} F_x^{cd} (u_1^c - u_1^d), \quad (5)$$

$$G_{III} = \lim_{\Delta a \rightarrow 0} \frac{1}{2b\Delta a} F_z^{cd} (u_3^c - u_3^d). \quad (6)$$

Thus, the total strain–energy release rate is

$$G = G_I + G_{II} + G_{III}. \quad (7)$$

For the present two–dimensional DCB configuration, only the opening mode contributes. The Mode I VCCT therefore reduces to

$$G_I = \lim_{\Delta a \rightarrow 0} \frac{1}{2b\Delta a} F_y^{cd} (u_2^c - u_2^d), \quad (8)$$

which is the formulation employed in this study to compute the Mode I strain–energy release rate for comparison with analytical and cohesive-zone predictions. Although a mixed–mode VCCT framework is retained in the numerical implementation for generality, the present simulations are restricted to Mode I loading through the prescribed boundary conditions. As a result, the shear and tearing contributions to the strain–energy release rate remain negligible throughout the analysis, and crack propagation is governed solely by G_I .

It is important to emphasize that VCCT is based on LEFM assumptions and therefore cannot capture progressive softening, nonlinear traction–separation behavior, and mixed–mode degradation in contrast to the CE and CS approaches evaluated later in this work.

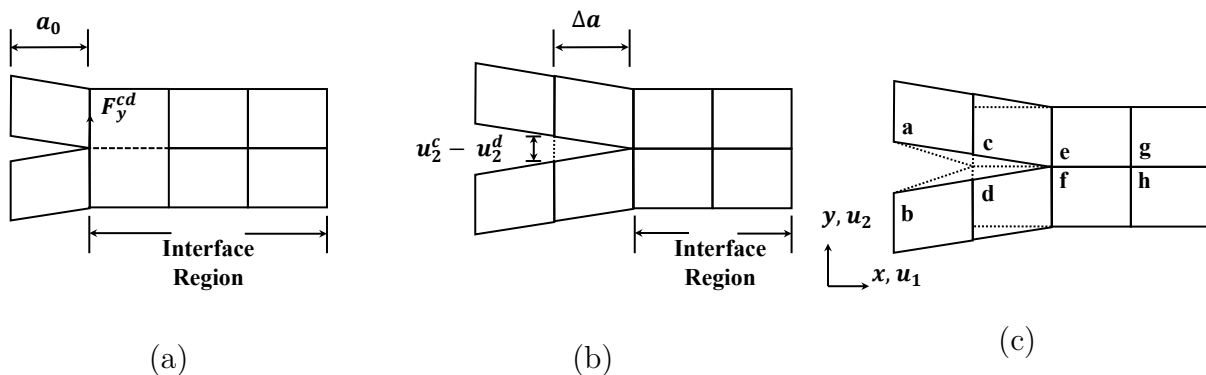


Figure 1: Schematic illustration of the VCCT. (a) Initial crack configuration with crack length a_0 , showing the vertical component of the nodal reaction force F_y^{cd} acting at the first node pair ahead of the crack tip. The interface region indicates the elements over which VCCT quantities are extracted. (b) Virtual crack extension Δa and the corresponding relative opening displacement $(u_2^c - u_2^d)$ measured behind the crack tip. (c) VCCT node–pairing arrangement showing the displacement pair behind the crack tip (a, b) and the corresponding force pair ahead of the crack tip (c, d) used in the Mode I energy release rate calculation. Nodes (e, f) and (g, h) denote additional interface nodes further ahead in the mesh and are shown only for completeness.

2.1.2 CZM

CZM provides a nonlinear fracture mechanics framework for representing delamination through a traction–separation constitutive relation defined along the interface. In contrast to the traction-free crack tip assumption of VCCT, CZM introduces a finite fracture process zone

in which tractions continue to transfer across partially separated crack faces. Early theoretical foundations of the cohesive zone model were established by Dugdale [26] and Barenblatt [7], who demonstrated that stresses at a crack tip do not drop instantaneously to zero but instead develop within a finite cohesive region. Building on these concepts, Hillerborg et al. formulated a traction–separation law for quasi-brittle fracture providing the basis for modern finite element implementations of CZM [41]. Two classes of CZM formulations are commonly employed [6]:

- **CE**, which possess a finite (usually small) thickness and are suited for modelling adhesive layers or interphase materials where the interface thickness is physically meaningful.
- **CS**, which have zero thickness and are inserted between coincident nodes of adjoining laminates. These surfaces transfer tractions until damage initiates and evolves.

In both formulations, the interface response is governed by a traction–separation law to represent progressive stiffness degradation in terms of the nominal traction vector $\mathbf{t} = \{t_n; t_s; t_t\}$ and the corresponding separation $\mathbf{\Delta} = \{\Delta_n; \Delta_s; \Delta_t\}$ with a scalar damage variable $d \in [0, 1]$ [55]. The simplest and most widely used model is the bilinear traction–separation law, illustrated in Figure 2 for Mode I loading. The fracture process consists of four stages: (1) linear elastic response, (2) damage initiation at the peak traction, (3) progressive softening, and (4) complete decohesion. The constitutive relation is written as

$$\mathbf{t} = (1 - d) \mathbf{K} \mathbf{\Delta}, \quad (9)$$

where $\mathbf{K} = \text{diag}(K_n, K_s, K_t)$ is the undamaged interface stiffness matrix. Prior to damage initiation, $d = 0$ and the response is linear elastic. Complete decohesion corresponds to $d = 1$, for which $\mathbf{t} = \mathbf{0}$.

Damage Initiation Criterion. In ABAQUS, interface damage initiation is typically governed by the quadratic nominal stress criterion (QUADS),

$$\left(\frac{\langle t_n \rangle}{t_n^0} \right)^2 + \left(\frac{t_s}{t_s^0} \right)^2 + \left(\frac{t_t}{t_t^0} \right)^2 = 1, \quad (10)$$

where t_n^0 , t_s^0 , and t_t^0 represent peak nominal tractions, and $\langle \cdot \rangle$ denotes the Macaulay bracket to exclude compressive normal traction from damage initiation. When Eq. (10) is first satisfied, damage initiates (d begins to evolve from zero). For convenience, an equivalent separation is introduced as the Euclidean norm of the separation vector [23], i.e. ,

$$\Delta_{\text{eq}} = \sqrt{\Delta_n^2 + \Delta_s^2 + \Delta_t^2}, \quad (11)$$

with an associated history variable $\bar{\Delta}_{\text{eq}}(t) = \max_{\tau \leq t} \Delta_{\text{eq}}(\tau)$ to ensure irreversibility of damage, i.e., once d increases it cannot decrease upon unloading or reloading.

Damage Evolution Law. After damage initiation, the damage variable evolves according to a bilinear softening law driven by the maximum equivalent separation $\bar{\Delta}_{\text{eq}}$ [23],

$$d = \begin{cases} 0, & \bar{\Delta}_{\text{eq}} \leq \Delta_{\text{eq}}^0, \\ \frac{\Delta_{\text{eq}}^f (\bar{\Delta}_{\text{eq}} - \Delta_{\text{eq}}^0)}{\bar{\Delta}_{\text{eq}} (\Delta_{\text{eq}}^f - \Delta_{\text{eq}}^0)}, & \Delta_{\text{eq}}^0 < \bar{\Delta}_{\text{eq}} < \Delta_{\text{eq}}^f, \\ 1, & \bar{\Delta}_{\text{eq}} \geq \Delta_{\text{eq}}^f, \end{cases} \quad (12)$$

where Δ_{eq}^0 denotes the equivalent separation at damage initiation (i.e., the state at which Eq. (10) is first satisfied), and Δ_{eq}^f is the equivalent separation at complete decohesion. For mixed-mode delamination, ABAQUS employs the Benzeggagh-Kenane (BK) law, which relates the critical fracture energy to the energy release rates in each mode. The general expression is

$$G_c = G_{Ic} + (G_{IIc} - G_{Ic}) \left(\frac{G_{II} + G_{III}}{G_I + G_{II} + G_{III}} \right)^\eta, \quad (13)$$

and damage growth occurs when the energy dissipated by interface separation reaches the critical fracture energy G_c . For a bilinear law, Δ_{eq}^f is chosen such that the area under the traction–separation response equals G_c .

For the present pure Mode I DCB configuration, $\Delta_s = \Delta_t = 0$ and $t_s = t_t = 0$. The QUADS criterion Eq.(10) reduces to

$$\frac{\langle t_n \rangle}{t_n^0} = 1. \quad (14)$$

Under pure Mode I loading, $\Delta_{\text{eq}} = |\Delta_n|$ and therefore $\Delta_{\text{eq}}^f = \Delta_{nc}$ and $\Delta_{\text{eq}}^0 = \Delta_{n1} = t_n^0/K_n$. The BK relation (13) simplifies to $G_c = G_{Ic}$. It is noted that a mixed-mode cohesive formulation is employed in all simulations to maintain consistency with the general CZM framework. However, for the present Mode I DCB configuration, the imposed boundary conditions ensure that only normal separation occurs, and the interface response reduces exactly to a pure Mode I traction–separation law. For the bilinear Mode I traction–separation law, the fracture energy equals the triangular area under the curve,

$$G_{Ic} = \frac{1}{2} t_n^0 \Delta_{nc}, \quad (15)$$

where Δ_{nc} is the normal separation at complete decohesion, which is fully consistent with the analytical DCB solution and the VCCT formulation employed in this work.

2.2 Analytical Model for Mode I Delamination

Figure 3(a) illustrates the Mode I DCB configuration considered in this study. The physical specimen consists of two identical laminate arms separated by an initial crack of length a_0 and loaded in symmetric opening by forces P . Because the two arms deform identically, the total strain energy of the full DCB is twice the strain energy stored in a single arm. For analytical modeling, it is therefore convenient to idealize the DCB as shown in Figure 3(b), where one arm is treated as an Euler–Bernoulli cantilever beam of physical crack length a .

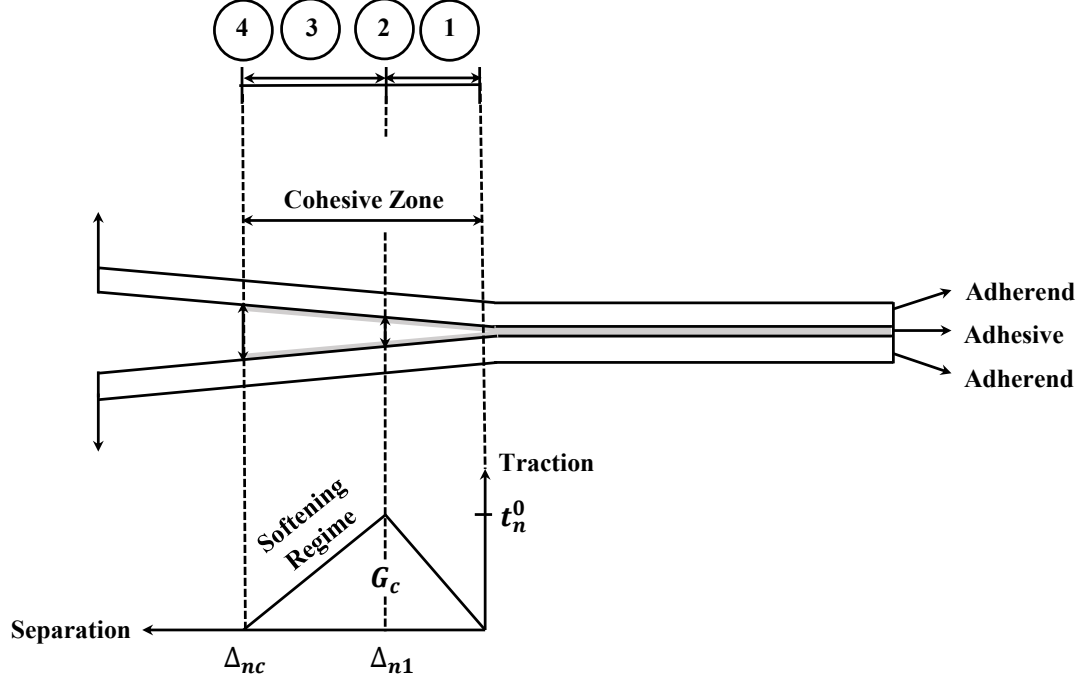


Figure 2: Schematic representation of the bilinear traction–separation cohesive zone model for Mode I delamination. The interface response consists of an initial linear elastic regime (1), damage initiation at peak traction t_n^0 (2), progressive softening governed by the traction–separation law (3), and complete decohesion at the critical separation Δ_{nc} (4). The area under the traction–separation curve corresponds to the cohesive fracture energy G_c .

The arm is fixed at the crack tip and loaded by a vertical force P at the free end. The analytical expressions derived using this single-arm idealization are then used to compute the global (full DCB specimen) force–displacement response. However, opening is imposed via prescribed end displacements in the finite element simulations.

The classical closed-form displacement of a DCB of effective length $\bar{a}$ is

$$\delta = \frac{4P\bar{a}^3}{E_1 B h^3}, \quad (16)$$

where E_1 is the longitudinal modulus of the adherends, B is the specimen width, and h is the full thickness of one arm (so the total DCB thickness is $2h$). Because the strain energy of the full DCB is twice that of a single arm, the resulting global opening displacement contains the factor of 4 in Eq. 16, consistent with standard DCB formulations.

The corresponding Mode I energy release rate for the two-arm DCB is

$$G_I = \frac{12P^2\bar{a}^2}{E_1 B^2 h^3}, \quad (17)$$

which follows from differentiating the total DCB strain energy (twice the single-arm energy) with respect to the applied load.

Enforcing the Griffith condition $G_I = G_{Ic}$ gives the critical (peak) load

$$P_c = \sqrt{\frac{G_{Ic} E_1 B^2 h^3}{12 \bar{a}^2}}. \quad (18)$$

The corresponding displacement follows from Eq. (16), $\delta_c = \frac{4P_c \bar{a}^3}{E_1 B h^3}$. Because the crack tip in an orthotropic laminate does not behave as a perfectly clamped root, a crack-tip rotation correction is introduced following the expressions of Williams [81] and Pereira & de Morais [65]. The effective crack length is $\bar{a} = a + \Delta_c$, with the correction

$$\Delta_c = h \sqrt{\frac{E_1}{11G_{12}} \left[3 - 2 \left(\frac{\Gamma}{1 + \Gamma} \right)^2 \right]}, \quad \Gamma = 1.18 \sqrt{\frac{E_1 E_2}{G_{12}}}. \quad (19)$$

Eqs. (16)-(18), together with the correction in Eq. (19), provide the analytical force—displacement curve that is used to validate the VCCT, CE, and CS results presented later. Prior to crack growth, the physical crack length is fixed at $a = a_0$, while the crack-tip rotation correction is incorporated through the effective length $\bar{a}_0 = a_0 + \Delta_c$. During crack propagation, the physical crack length a increases, and the same correction is applied via $\bar{a} = a + \Delta_c$. A detailed derivation used to generate the analytical curves are provided in Appendix B. The analytical solution is derived within the framework of linear elastic fracture mechanics and is therefore strictly valid under small-strain, linear elastic conditions. Since VCCT is itself based on LEFM and CZM recovers the LEFM limit under sufficiently high cohesive stiffness and small process-zone conditions, meaningful comparison with the analytical solution is ensured.

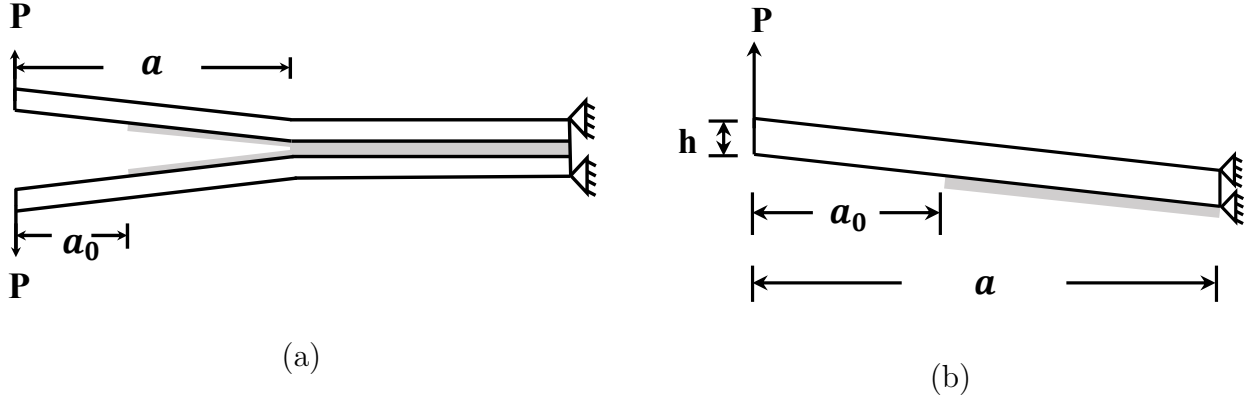


Figure 3: (a) Geometry of the DCB specimen under Mode I loading, showing the initial crack length a_0 . (b) Corresponding analytical single-arm representation, in which the physical crack length a defines the cantilever span.

2.3 Finite Element Modeling Framework

A two-dimensional DCB configuration is analyzed to investigate Mode I delamination using three fracture modeling strategies: VCCT, CE, and CS. The objective is to enable a direct

and consistent comparison among these approaches under identical geometric, material, and loading conditions. As discussed in the analytical formulation section 2.2, the DCB response is derived analytically using a single-arm cantilever idealization subjected to an opening force. In the finite element simulations, the same Mode I opening is imposed through symmetric prescribed displacements at the loading end, while the opposite end remains traction-free, as shown in Figure 4. Although mixed-mode interface properties are specified for completeness in the ABAQUS definitions, the symmetric vertical displacement loading produces a strictly Mode I opening response in all simulations considered in this work. This displacement-controlled implementation is commonly adopted in numerical DCB simulations for improved numerical stability during crack growth and yields an equivalent global force-displacement response, enabling direct comparison with analytical solutions [10, 52, 79].

The DCB specimen has a total length of 100 mm and a width of 20 mm. Each arm has a thickness of 1.5 mm, resulting in a total specimen thickness of 3 mm. An initial delamination of length $a_0 = 30$ mm is introduced along the mid-plane, as illustrated in Figure 4. The adherends are modeled as orthotropic CFRP composite laminates, while the interface is represented using either CE, CS interactions, or a conventional continuum interface for VCCT simulations.

The composite arms are discretized using four-noded bilinear plane-strain elements with incompatible modes (CPE4I), which provide improved accuracy for bending-dominated deformation [6]. A direct comparison with plane-stress elements (CPS4I) shows that the predicted peak load differs by less than 2% (Appendix C), indicating negligible sensitivity of the global response to the two-dimensional modeling assumption. The CE are discretized using zero-thickness interface elements with a bilinear traction-separation law. For the CS formulation, the same traction-separation parameters are enforced through surface-based cohesive interactions to ensure a consistent comparison with the CE approach. In the VCCT simulations, the interface is modeled using standard continuum elements and the strain-energy release rate is computed from nodal force-displacement pairs at the crack tip.

Material properties of the composite adherends are summarized in Table 1 [50], while the cohesive zone parameters used for both CE and CS are listed in Table 2 [50]. Identical cohesive properties are employed across all simulations to isolate the effect of the fracture modeling formulation.

Table 1: Orthotropic elastic properties of the composite adherends.

Property	Value
E_1 (GPa)	122.7
$E_2 = E_3$ (GPa)	10.1
$\nu_{12} = \nu_{13}$	0.25
ν_{23}	0.45
$G_{12} = G_{13}$ (GPa)	5.5
G_{23} (GPa)	3.7

Loading is applied by prescribing symmetric vertical displacements at the left end of the specimen. The upper arm is displaced upward by $u_2 = +8$ mm and the lower arm

Table 2: Cohesive zone model parameters used for CE and CS formulations.

Parameter	Value
Peak normal traction, t_n^0 (MPa)	80
Peak shear tractions, $t_s^0 = t_t^0$ (MPa)	80
Mode-I fracture energy, G_{Ic} (J/m ²)	969
Mode-II fracture energy, G_{IIc} (J/m ²)	969
Mode-III fracture energy, G_{IIIc} (J/m ²)	969
BK exponent	2.284
Penalty stiffness (N/m ³)	8.0×10^{14}
Viscosity (s)	1.0×10^{-5}

downward by $u_2 = -8$ mm. Both loading points are constrained in the horizontal direction ($u_1 = 0$) to suppress rigid-body motion. No boundary conditions are applied at the right end of the DCB, allowing it to remain traction-free. This displacement-controlled configuration produces pure Mode I opening while avoiding artificial bending constraints at the crack front, thereby facilitating a consistent comparison among the VCCT, CE, and CS formulations. Reaction forces and displacements at the loading points are recorded to construct global load-displacement curves, which are subsequently compared against analytical predictions and among the three fracture modeling approaches.

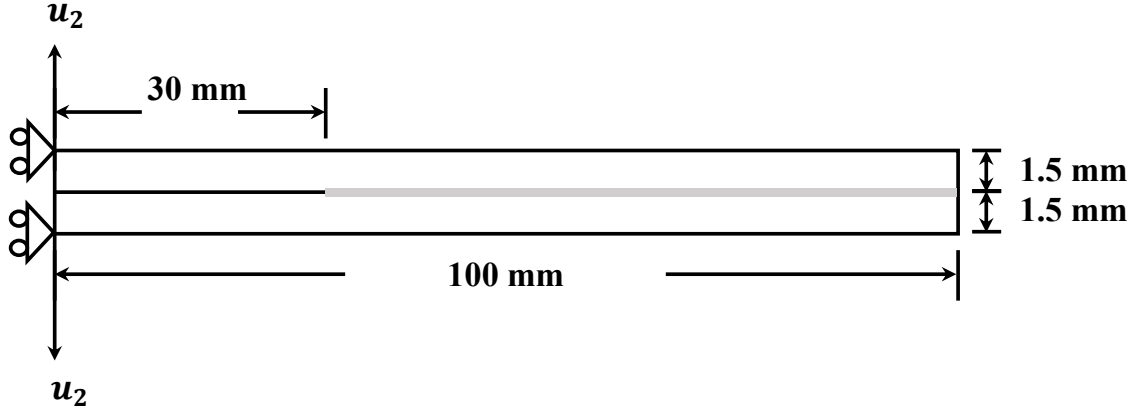


Figure 4: Geometry and boundary conditions of the DCB specimen used in the finite element simulations.

3 Results and discussion

This section outlines the evaluation and discussion of the numerical results derived from the simulations. The study evaluates the mesh sensitivity of three delamination modeling techniques (VCCT, CE, and CS) under Mode I loading using a DCB configuration. The convergence behavior is assessed through load-displacement responses, critical (peak) load

trends, and mesh-dependence characteristics. Computational time and fiber orientation effects are also investigated.

3.1 Mesh Convergence Analysis

A systematic mesh convergence study is performed for the three delamination modeling approaches considered in this work: VCCT, CE, and CS. While the mesh sensitivity of fracture simulations is well recognized, the degree and mechanism of this sensitivity differ substantially among these formulations. The objectives of this section are therefore twofold: (i) to quantify the level of interface discretization required for each method to achieve mesh-independent predictions of the Mode I critical load in a DCB configuration, and (ii) to demonstrate that all three approaches converge to a consistent physical response when adequately refined. For all simulations, the composite arms are discretized using CPE4I plane-strain elements, while the interface is uniformly meshed along the beam length. Both the bulk mesh and the interface discretization are refined simultaneously to ensure consistent resolution of the deformation field in the composite arms and the traction-separation response along the interface. For the VCCT and CS, the reported element counts refer to the discretization of the interface along the beam length with the bulk mesh refined proportionally. In contrast, the CE employs the same bulk discretization but introduces additional cohesive elements along the interface, resulting in a larger total number of elements and degrees of freedom for a given interface resolution. The refinement strategy preserves mesh compatibility across the interface, such that changes in the predicted peak load reflect the combined effect of bulk and interface resolution. The number of interface elements is varied from 100 to 3200 using uniform refinement, corresponding to element lengths of 3.0, 2.0, 1.0, 0.3, and 0.25 mm. The relative refinement between the bulk elements and the interface discretization is maintained across all mesh levels to avoid spurious stiffness effects associated with incompatible length scales. The peak load extracted from the load-displacement response is adopted as the convergence metric. Convergence is evaluated based on the global peak load response, as this quantity governs the onset of delamination in DCB configurations and is the primary metric of interest in Mode I fracture analyses. Relative errors reported in Table 3 are computed with respect to the 2400-element mesh, which lies near the center of the identified convergence plateau and differs from the finest (3200-element) discretization by less than 1.7 % for all formulations, while reducing the computational cost. The 2400-element mesh is therefore used in all subsequent analyses. In addition, the converged numerical results are compared against the analytical Mode I peak load (~ 113.81 N) to assess the accuracy of each formulation (Table 4).

Among the three methods, VCCT exhibits the weakest dependence on interface mesh refinement. As shown in Figure 5(a), the predicted critical (peak) load varies only modestly across the full range of discretizations. The coarsest mesh (100 elements) yields a peak load of 115.25 N, while the finest mesh (3200 elements) predicts 116.74 N. The maximum variation of critical load across all meshes, computed as $(p_{\max} - p_{\min})/p_{2400}$, remains approximately 2.7% and the three finest discretizations (1600–3200 elements) differ by less than 1.4%. This behavior reflects the intrinsic stability of the VCCT formulation in line with prior VCCT studies [52, 60, 82], i.e. the energy release rate is computed directly from nodal forces and relative displacements at the crack tip without invoking a traction-separation law or damage

evolution. Consequently, VCCT requires only moderate mesh refinement to achieve stable predictions and therefore provides a useful baseline for evaluating the cohesive formulations.

In contrast, the CE formulation exhibits pronounced mesh dependence at coarse discretizations. As illustrated in Figure 5(b), the critical (peak) load is severely overpredicted for the 100 and 400 element meshes, reaching 313.40 N and 161.77 N, respectively. These values lie far above the physically expected range and result from the inability of coarse meshes to resolve the fracture process zone [1, 9]. Under such conditions, the discrete interface behaves overly stiff as a result of the large element lengths, delaying damage initiation and artificially increasing the apparent interface stiffness. Upon refinement to 1600 elements, the predicted peak load decreases sharply to 112.64 N and remains essentially unchanged with further refinement (112.83 N at 2400 elements and 112.35 N at 3200 elements). The variation among the three finest meshes is below 0.5%, indicating that approximately 1600 or more interface elements are required for mesh-independent CE predictions in the present configuration.

The CS formulation displays convergence characteristics similar to those of CE, but with slightly larger variability in the converged regime. As shown in Figure 5(c), coarse meshes again lead to substantial overprediction of the critical load (318.67 N for 100 elements and 158.89 N for 400 elements), reflecting inadequate resolution of the fracture process zone. Refinement to 1600 elements reduces the peak load to 114.26 N, and subsequent refinement produces only minor changes (115.67 N at 2400 elements and 113.76 N at 3200 elements). The variation among the three finest discretizations remains below 1.7%, demonstrating satisfactory convergence. Compared to CE, the slightly larger scatter observed for CS is consistent with its surface-based implementation [2]. In CS, the cohesive tractions are evaluated directly from the kinematics of the adjacent bulk elements, which can introduce marginally higher numerical sensitivity to interpolation and mesh-alignment effects than discrete cohesive elements. For both cohesive (CE and CS) formulations, the same traction-separation law parameters are used for all mesh densities. The cohesive strength is $t_n^0 = 80 \times 10^6$ Pa, the fracture energy is $G_{Ic} = 969$ J/m², and the penalty stiffnesses are prescribed as $K_{nn} = K_{ss} = K_{tt} = 8 \times 10^{14}$ N/m³. These values are intentionally kept constant during mesh refinement because the damage-initiation displacement,

$$\Delta_{n1} = \frac{t_n^0}{K_n} \approx 1 \times 10^{-4} \text{ mm},$$

is very small. Thus, mesh refinement primarily affects the ability to resolve the fracture process zone and the associated traction gradients, while the underlying traction-separation law parameters remain unchanged. The observed convergence behavior for the cohesive formulations indicates that the chosen interface discretizations adequately resolve the cohesive process zone length, beyond which further mesh refinement has negligible influence on the peak load. This practice of keeping the traction-separation parameters fixed while refining the interface mesh is widely adopted in CZM-based DCB delamination simulations, provided convergence is deemed achieved when the change in peak load between successive refinements falls below approximately 1–2% [61, 74].

To further evaluate the accuracy of the converged solutions, the numerical peak loads obtained using the 2400-element mesh are compared with the analytical Mode I solution in Table 4. All three numerical approaches show close agreement with the analytical prediction.

Among them, the CE formulation exhibits the smallest deviation from the analytical peak load, while the CS and VCCT formulations exhibit a small overprediction of the analytical value. The modest overprediction observed for VCCT is consistent with its force-based evaluation of the energy release rate [52, 60, 82]. Overall, all three methods converge to a consistent prediction of the Mode I critical (peak) load within a narrow range of 112–116 N when sufficient mesh refinement is employed. However, the required level of refinement differs markedly among the formulations [60]. VCCT achieves stable results even on relatively coarse meshes, whereas CE and CS require approximately 1600 or more interface elements to accurately resolve the fracture process zone and eliminate mesh-induced stiffness effects. This systematic comparison provides quantitative guidance on the mesh requirements and accuracy trade-offs of commonly used delamination modeling strategies under identical loading and material conditions.

Table 3: Peak load and relative error for each mesh discretization. All errors are computed with respect to the 2400-element mesh used in the subsequent analyses.

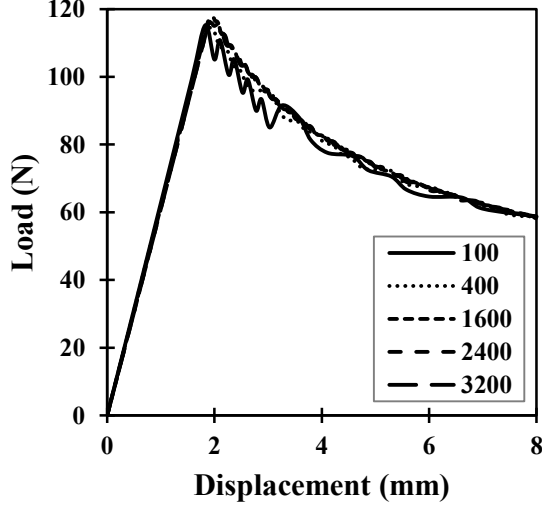
Elements	VCCT		CE		CS	
	Peak (N)	Error (%)	Peak (N)	Error (%)	Peak (N)	Error (%)
100	115.25	0.59	313.40	177.76	318.67	175.50
400	114.38	1.33	161.77	43.37	158.89	37.36
1600	117.53	1.38	112.64	0.17	114.26	1.22
2400	115.93	0.00	112.83	0.00	115.67	0.00
3200	116.74	0.70	112.35	0.43	113.76	1.65

Table 4: Comparison of converged numerical peak loads with the analytical Mode I solution. Numerical results correspond to the 2400-element mesh used in subsequent analyses.

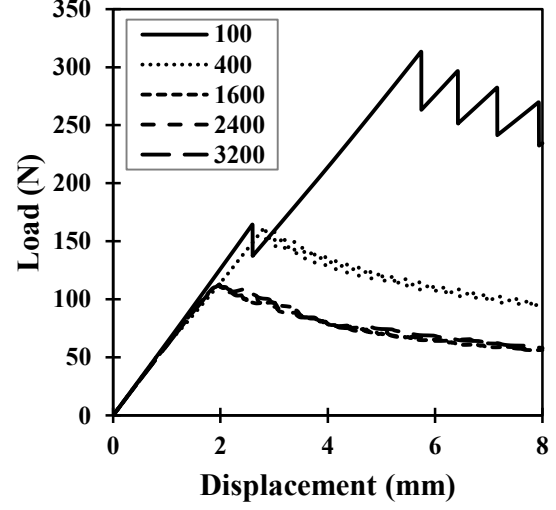
Method	Elements	Peak Load (N)
VCCT	2400	115.93
CE	2400	112.83
CS	2400	115.67
Analytical	-	113.81

3.2 Comparison of load–displacement response

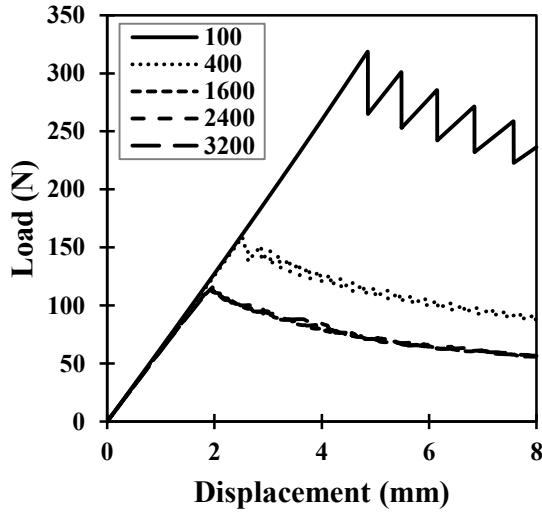
Figure 6(a) compares the load–displacement responses obtained using the VCCT, CE, CS, and the benchmark analytical solution for the Mode I DCB configuration (Section 2.2 and Appendix B). In the initial loading stage, all numerical approaches exhibit an almost identical linear slope and closely follow the analytical prediction, indicating that the global compliance of the specimen is governed by elastic beam bending and is independent of the delamination formulation prior to crack initiation [56, 85]. This agreement confirms that none of the interface models introduce artificial stiffness in the pre-damage regime.



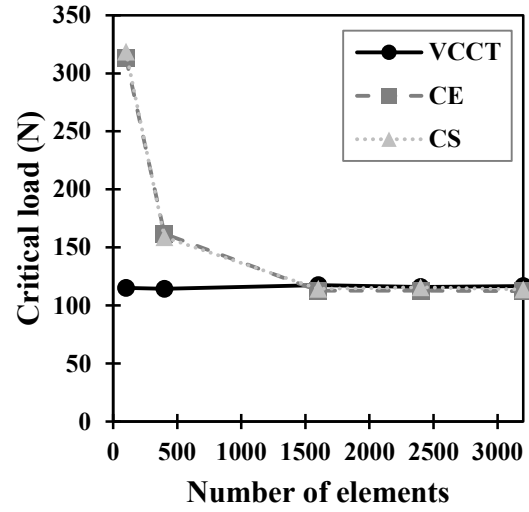
(a)



(b)



(c)



(d)

Figure 5: Mesh-dependent load–displacement responses for (a) VCCT, (b) CE, and (c) CS, together with (d) the corresponding convergence of the peak load for all three formulations.

As the applied displacement increases, deviation from linearity marks the onset of delamination and the attainment of the peak load. The predicted critical loads from the VCCT, CE, and CS models show close agreement with the analytical solution (Figure 6 and Table 4), demonstrating that all three numerical approaches accurately reproduce delamination initiation under linear elastic Mode I loading conditions when appropriate fracture parameters are prescribed [39, 74]. This correspondence provides validation of the numerical implementations against a closed-form analytical benchmark.

Differences among the methods become more evident in the post-peak regime. The VCCT response exhibits a comparatively smooth reduction in load following crack initiation, reflecting the absence of an explicitly modeled fracture process zone and the assumption of a traction-free crack tip once the critical energy release rate is exceeded [17]. In contrast, CE and CS formulations display a more fluctuating softening response, which arises from progressive damage evolution within a finite fracture process zone governed by the traction–separation law [63, 74]. Localized stiffness degradation and sensitivity to incremental separation lead to small oscillations in the predicted load, particularly during crack propagation [38, 40, 56].

To further quantify the agreement with the analytical benchmark, Figure 6(b) presents the relative error between each numerical prediction and the analytical solution as a function of displacement over the common range shared by all methods. The relative error is defined as

$$e = \frac{|P_{\text{num}} - P_{\text{ana}}|}{\max(|P_{\text{ana}}|, 0.01 P_{\text{ana}}^{\text{max}})} \times 100\%,$$

where the loads are evaluated at the same displacement and the lower bound prevents numerical singularities at small load values. All approaches exhibit low error levels in the elastic regime and in the vicinity of the critical load, confirming accurate prediction of stiffness and delamination initiation [16, 39]. Increased fluctuations in relative error are observed in the post-peak region, where the numerical models capture progressive damage evolution and crack propagation mechanisms that are not represented in the idealized analytical solution [56, 63]. These deviations therefore reflect differences in fracture representation rather than numerical inconsistency.

It is noted that the bilinear traction–separation law employed in this study does not account for fiber-bridging effects, which are known to influence the post-peak response of composite laminates [72, 83]. While more advanced trilinear or multilinear cohesive laws can incorporate such mechanisms [34, 72, 83], the present comparison focuses on commonly used delamination formulations under consistent assumptions to assess their agreement with an analytical benchmark.

3.3 Energy balance in VCCT and CZM

The global energy response of the delamination simulations is examined by comparing the external work and internal energy obtained from Abaqus history outputs. The external work corresponds to the work done by the imposed displacement boundary condition (ALLWK), while the internal energy represents the energy stored within the system due to deformation and damage processes (ALLIE). It is noted that the internal energy reported by Abaqus (ALLIE) represents elastic strain energy for VCCT, while for cohesive zone formulations it additionally includes energy dissipated through damage evolution at the interface [21, 22]. The distinct energy trends observed in Figure 7(a)–(c) for VCCT and cohesive zone formulations arise from their fundamentally different fracture representations. In the VCCT framework, delamination is governed by an energy release criterion derived from linear elastic fracture mechanics, in which crack growth is controlled by the energy release rate $G_I = -\partial\Pi/\partial a$, evaluated from nodal forces and displacements [82]. Since VCCT does not introduce a constitutive damage mechanism or traction–separation law at the interface, the energy required

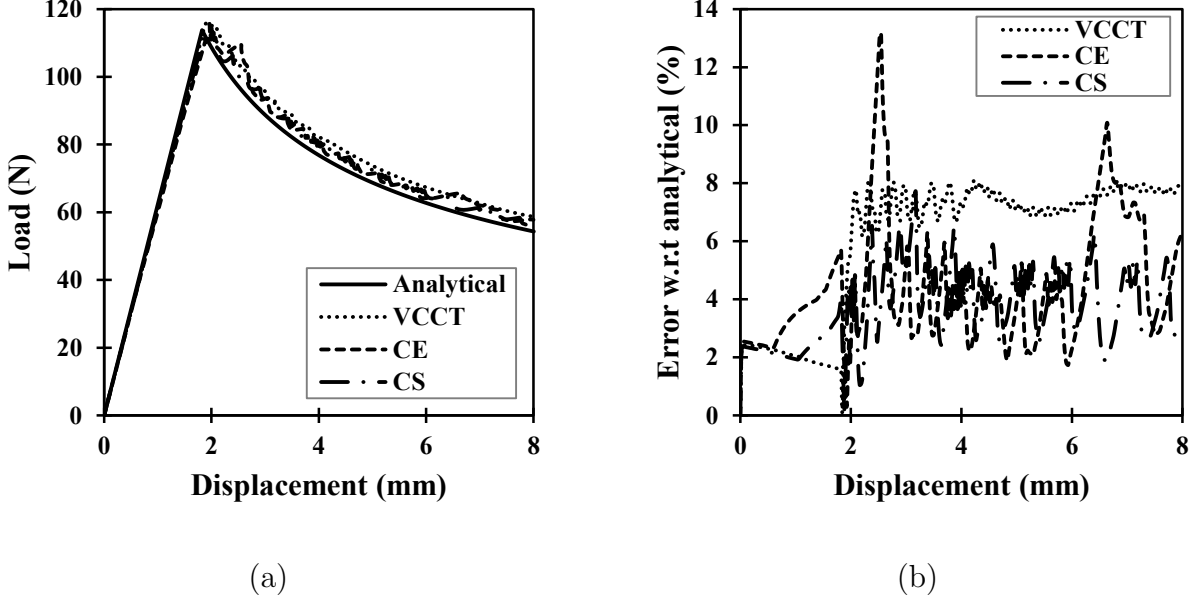


Figure 6: (a) Comparison of load–displacement responses obtained using VCCT, CE, CS, and the analytical solution for the Mode I DCB configuration. (b) Relative error between numerical predictions and the analytical solution as a function of displacement, evaluated over the common displacement range shared by all methods.

to advance the crack is released from the system rather than dissipated internally [78]. In contrast, CE and CS formulations model delamination through a traction–separation law, where the fracture energy G_{Ic} is given by the area under the traction–separation curve and is dissipated internally during damage evolution [64].

For the VCCT formulation, the external work and internal energy coincide during the initial loading stage, indicating a purely elastic response prior to crack growth. Once delamination initiates, the two curves begin to diverge, as shown in Figure 7(a). The increase in external work exceeds the increase in internal energy because a portion of the applied energy is expended in creating new crack surfaces. In the VCCT approach, this energy is not stored as internal energy but is implicitly accounted for through the energy release rate used to drive crack propagation. Consequently, the separation between ALLWK and ALLIE provides a clear indication of energy expended during delamination growth [43, 53, 62]. For the present two–dimensional DCB configuration, the energy associated with crack advance is given by $G_I b \Delta a$, which corresponds to the difference between the external work (ALLWK) and internal energy (ALLIE) observed in the VCCT simulations [78]. At the final applied displacement of 8 mm, the VCCT simulation yields $ALLWK = 1.18$ J and $ALLIE = 0.47$ J, indicating that approximately 0.71 J of the applied work is associated with crack advance.

In contrast, for both CE and CS formulations, the external work and internal energy remain nearly identical throughout the loading process, as illustrated in Figures 7(b) and 7(c), with ALLWK and ALLIE remaining equal at approximately 1.15 J at the final applied displacement of 8 mm. In cohesive approaches, damage initiation and evolution are governed by a traction–separation law, and the energy associated with crack growth is incorporated

directly into the constitutive response of the interface. As a result, the energy dissipated due to delamination is included within the internal energy measure, leading to overlapping ALLWK and ALLIE curves [29, 64, 76]. This behavior reflects that fracture energy dissipation is embedded within the cohesive interface response rather than evaluated as an external energy release [64]. Although CS formulations are implemented within a contact framework, the dominant fracture energy dissipation in the present simulations arises from the cohesive traction–separation law, resulting in energy trends comparable to CE.

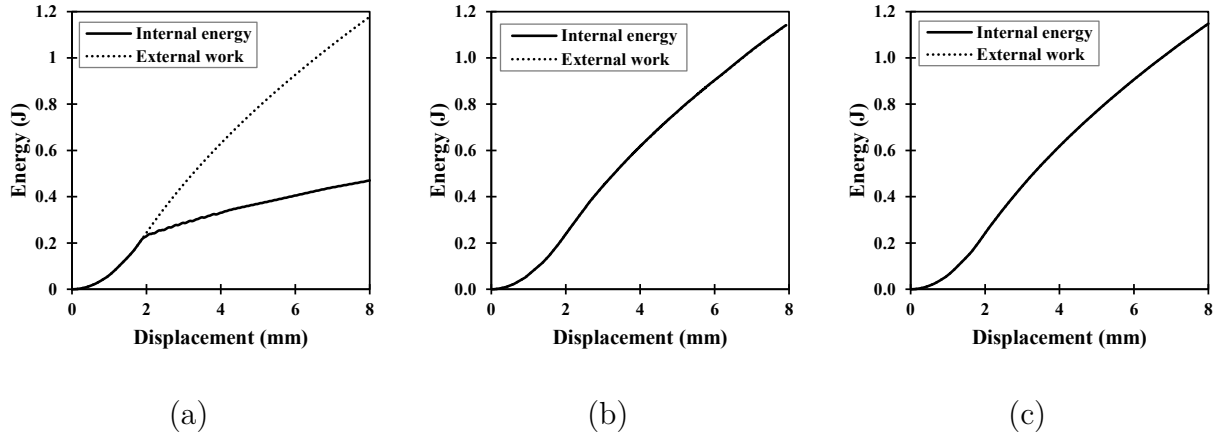


Figure 7: Energy consistency of (a) VCCT, (b) CE, and (c) CS, showing the evolution of external work (ALLWK) and internal energy (ALLIE) with applied displacement. In the VCCT formulation, a separation between ALLWK and ALLIE is observed after crack initiation due to the release of fracture energy associated with crack advance. In contrast, for CE and CS formulations, the fracture energy is dissipated through the traction–separation law and is therefore included within the internal energy, leading to overlapping ALLWK and ALLIE curves.

3.4 Crack length variation with CMOD

In fracture mechanics, the crack mouth opening displacement (CMOD) is widely used as a global measure of crack initiation and growth in Mode I double cantilever beam (DCB) tests. As the applied opening displacement increases, the interfacial crack extends progressively, as reflected by the increasing CMOD. Numerous experimental and numerical studies have demonstrated a strong correlation between CMOD and crack length, often exhibiting a near-linear or piecewise-linear relationship, particularly under stable Mode I delamination conditions [85].

In the present study, a consistent relationship between crack length and CMOD is observed for all three delamination modeling approaches considered: VCCT, CE, and CS. Figure 8(a) shows that the crack length initially remains nearly constant during the early loading stage, corresponding to stable opening of the pre-existing delamination prior to crack propagation [11]. This stage is followed by a more rapid increase in crack length as delamination propagates along the interface with increasing CMOD.

Although minor differences in the crack growth rate are observed among the three formulations, the overall crack length evolution with respect to CMOD remains in close agreement. This result is nontrivial, given the fundamentally different mechanisms used to model delamination in VCCT and cohesive zone formulations. VCCT assumes a traction-free crack tip and enforces crack growth through an energy release rate criterion, whereas the CE and CS approaches represent crack propagation through a distributed traction–separation law and progressive damage evolution along the interface. The close agreement among VCCT, CE, and CS predictions therefore demonstrates that, despite these underlying modeling differences [39], all three approaches recover a consistent global crack growth response when CMOD is used to characterize delamination progression in Mode I DCB configurations. This comparison provides a unified perspective on crack growth characterization across energy-based and traction-based delamination models, reinforcing the robustness of CMOD as a global descriptor for Mode I delamination.

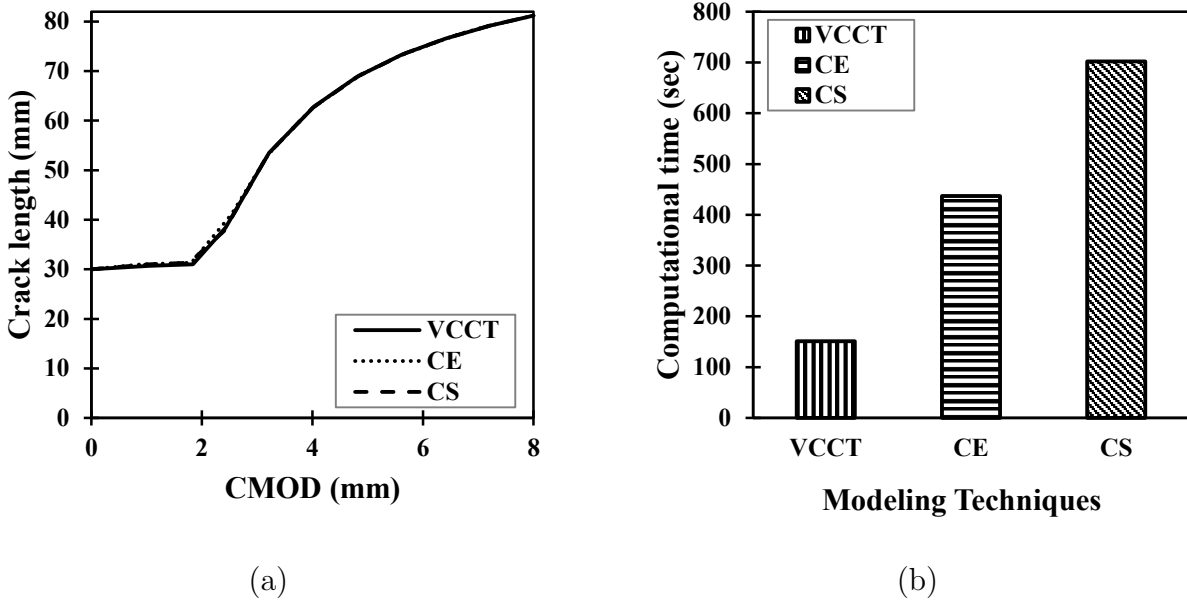


Figure 8: Panel (a) shows the corresponding crack length evolution as a function of CMOD for all three formulations. (b) Comparison of computational time required for delamination simulations using VCCT, CE, and CS formulations.

3.5 Computational efficiency of VCCT and CZM

In addition to predictive accuracy and fracture evolution characteristics, the computational efficiency of each delamination modeling approach is a critical consideration for practical applications and large-scale parametric studies [39]. The computational cost associated with the delamination simulations differs markedly among the three approaches considered [25, 66]. For identical mesh resolution, loading protocol, and convergence tolerances, the VCCT required approximately 151 s of wall-clock time to complete the analysis. In comparison, the CE formulation required 437 s, while the CS formulation exhibited the highest computational

cost, requiring 702 s, as summarized in Figure 8(b). All simulations were performed on the same hardware platform (Intel Core i5 processor with 8 GB system memory) using the default solver settings with single-core execution, ensuring a consistent and fair basis for comparison.

The lower computational cost of VCCT can be attributed to its relatively simple formulation, which does not introduce additional interface degrees of freedom or require nonlinear traction–separation constitutive updates prior to crack propagation [16, 25]. In contrast, the CE approach introduces explicit interface elements and history-dependent internal state variables associated with damage evolution, thereby increasing the complexity of local constitutive updates and the cost of nonlinear solution iterations [71]. Although the CS formulation avoids explicit interface discretization, it relies on contact-based constraint enforcement along the interface, leading to increased computational effort associated with contact detection, constraint enforcement, and iterative convergence [18].

Overall, the CS approach required approximately 4.6 times the computational time of VCCT for the present problem. While this additional cost may be justified in applications requiring enhanced modeling flexibility or robustness under complex contact and separation conditions, the results indicate that VCCT remains the most computationally efficient option for large-scale parametric studies or scenarios requiring repeated simulations. Although absolute computational times are hardware-dependent, the relative trends observed here reflect the intrinsic algorithmic complexity of each formulation.

3.6 Effect of ply orientation on delamination

The influence of ply orientation on Mode I delamination behavior is assessed for the DCB configuration by considering layups ranging from $0^\circ/0^\circ$ to $90^\circ/90^\circ$ [3, 8, 15]. To ensure that the observed trends reflect the underlying laminate anisotropy rather than modeling artifacts, the orientation study is performed using all three delamination formulations examined in this work: VCCT, CE, and CS [39]. Figure 9 compares the load–displacement response at each orientation. Across all layups, the three formulations yield nearly coincident curves, indicating strong consistency in both the pre-initiation stiffness and the post-initiation softening response. This agreement demonstrates that the orientation sensitivity reported here is largely formulation-independent, despite the fundamentally different fracture representations employed by VCCT (energy release rate) and cohesive zone approaches (traction–separation dissipation).

A clear and systematic effect of orientation is observed: as the ply orientation increases from 0° to 90° , the response becomes progressively more compliant, with (i) a reduction in the critical load and (ii) an increase in the displacement required to initiate and propagate delamination [3, 73]. These trends are quantified in Figure 10, which summarizes the critical load and critical displacement as functions of orientation. The critical load decreases sharply from 0° to 30° and approaches an approximately steady value for orientations above $\sim 60^\circ$, while the critical displacement increases monotonically over the same range. The observed decrease in critical load and increase in initiation displacement with increasing interface ply angle are consistent with experimentally reported orientation effects in Mode I delamination [4, 15]. Mechanistically, increasing orientation shifts the laminate response away from longitudinal fiber-dominated stiffness toward a more transverse-dominated compliance,

thereby reducing load-carrying capacity while requiring larger global deformation to deliver sufficient energy to the crack tip [73, 80].

From a design perspective, intermediate orientations (e.g., 30° – 60°) provide a useful trade-off between load capacity and deformation at delamination onset [42]. For example, a $45^\circ/45^\circ$ layup exhibits an intermediate response between the stiff $0^\circ/0^\circ$ configuration and the highly compliant $90^\circ/90^\circ$ case. While higher orientations (60° to 90°) delay delamination in terms of displacement, they do so primarily through increased compliance and do not necessarily imply improved interlaminar strength [67, 73]. Overall, the close agreement among VCCT, CE, and CS across the full orientation sweep provides a robust, formulation-independent benchmark for orientation effects in DCB delamination and supports the use of the present results as a reference dataset for future comparisons [39].

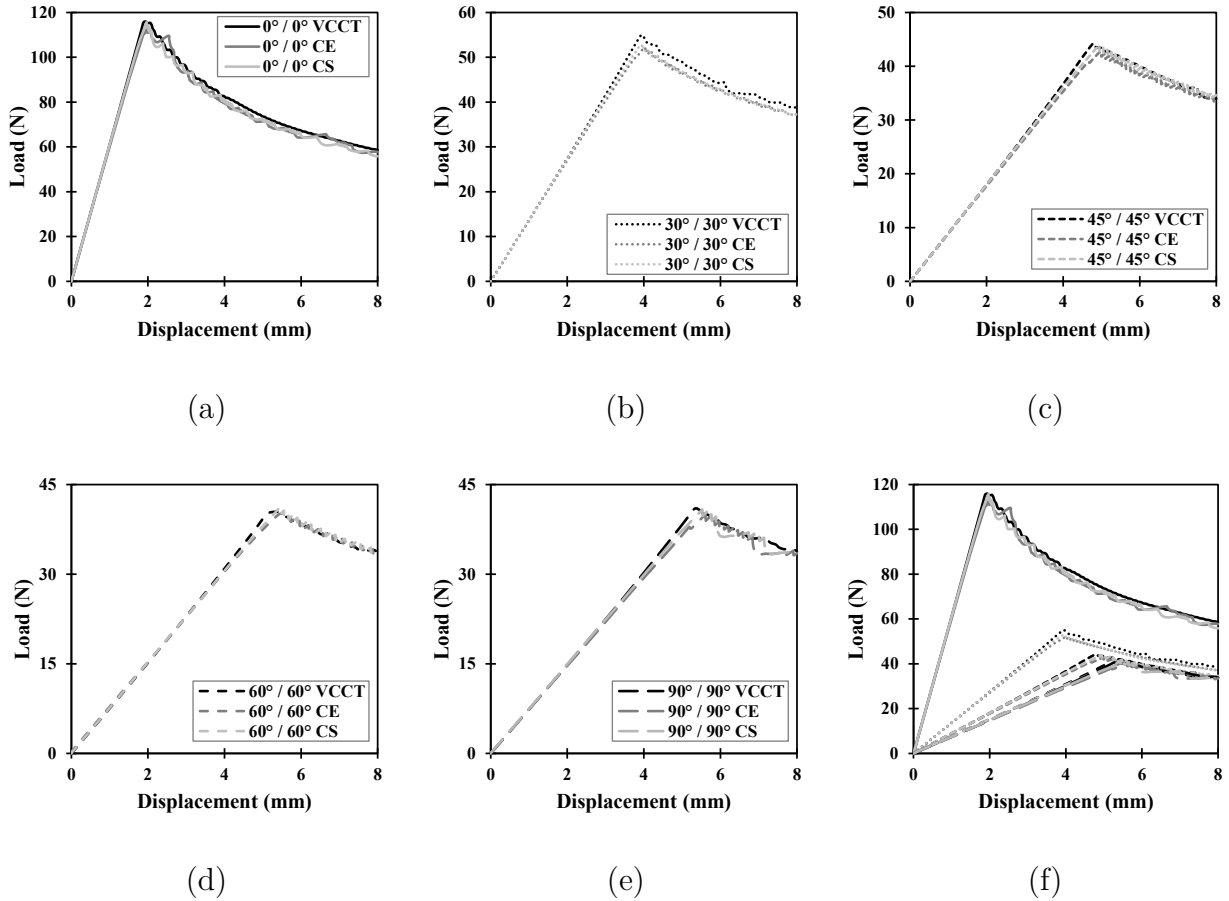


Figure 9: Load–displacement curves obtained using VCCT, CE, and CS for (a) $0^\circ/0^\circ$, (b) $30^\circ/30^\circ$, (c) $45^\circ/45^\circ$, (d) $60^\circ/60^\circ$, and (e) $90^\circ/90^\circ$ ply orientations. Panel (f) compares the responses for all orientations, highlighting the systematic reduction in peak load and the increase in displacement at delamination initiation with increasing ply orientation. The close agreement among the three formulations across all orientations demonstrates formulation-independent prediction of orientation effects.

Taken together, these results provide a unified and formulation-consistent benchmark

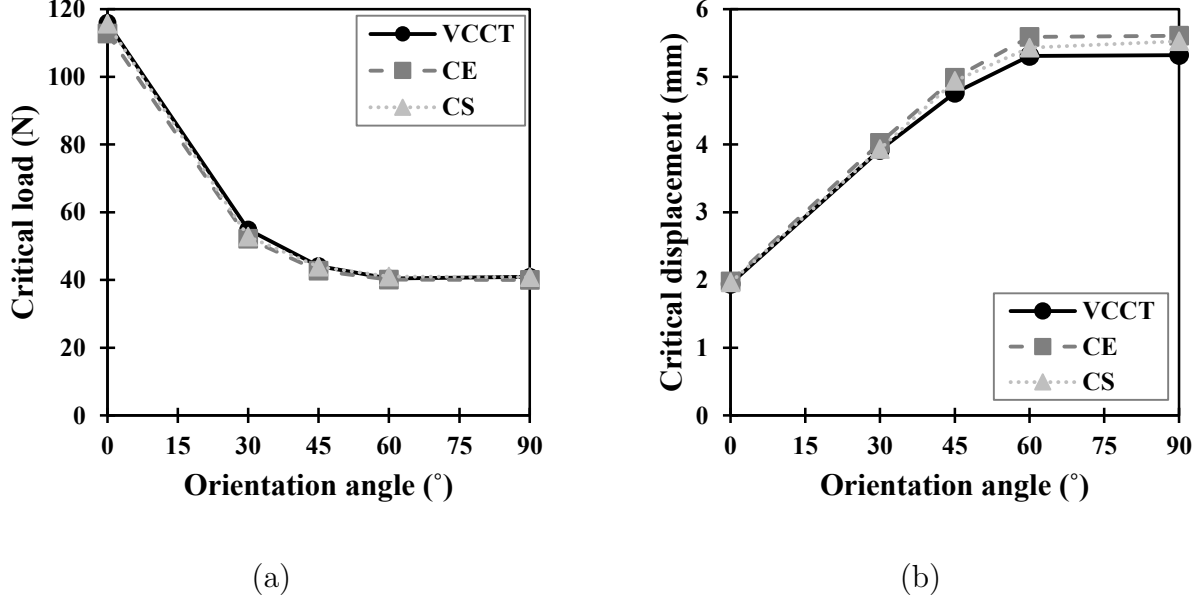


Figure 10: (a) Critical load and (b) critical displacement as functions of ply orientation angle predicted using VCCT, CE, and CS. Increasing orientation reduces the load required for delamination while increasing the displacement at initiation, reflecting increased laminate compliance.

for Mode I DCB delamination modeling. By evaluating VCCT, CE, and CS under identical geometry, material properties, fracture parameters, mesh refinement strategies, and solver settings, the present study isolates formulation-driven differences rather than artifacts of inconsistent numerical setup. The systematic comparison of mesh convergence behavior, energy balance characteristics, crack growth evolution, computational efficiency, and ply-orientation sensitivity provides quantitative guidance on mesh requirements and accuracy–cost trade-offs for commonly used delamination approaches.

4 Conclusions

This work has presented a systematic numerical investigation of Mode I delamination in laminated composite structures using three widely adopted modeling approaches: VCCT, CE, and CS. By maintaining consistent geometry, material properties, and loading conditions, the study has enabled a direct and fair comparison of their predictive accuracy, mesh sensitivity, energy characteristics, and computational efficiency.

The results demonstrate that VCCT exhibits the lowest mesh sensitivity and achieves stable, mesh-independent predictions with relatively coarse discretizations, making it particularly attractive for efficient delamination analyses when the crack path is known a priori. In contrast, cohesive-based formulations require significantly finer interface discretization to accurately resolve the traction–separation response, reflecting the presence of a finite fracture process zone. Despite these differences, all three approaches produce consistent

load–displacement responses at delamination onset and capture comparable crack length evolution when sufficient mesh refinement is employed.

Energy analyses further highlight fundamental distinctions among the formulations. In VCCT, fracture energy dissipation is evaluated explicitly during crack advance, leading to a corresponding reduction in stored elastic energy. For cohesive formulations, damage-induced dissipation is embedded within the internal energy response, resulting in external and internal energy histories that remain closely aligned throughout the simulation. From a computational standpoint, VCCT is the most efficient approach, followed by CE and CS, which incur higher costs due to additional degrees of freedom and nonlinear interface constitutive updates.

The influence of ply orientation on delamination behavior is also quantified. Laminates with 0° plies exhibit the highest load-carrying capacity under displacement-controlled loading, while 90° plies display increased crack opening with reduced tensile stiffness. Intermediate orientations provide a compromise between delamination resistance and structural compliance, underscoring the importance of stacking sequence design in damage-tolerant composite structures.

Overall, the study clarifies the strengths, limitations, and appropriate application regimes of VCCT and cohesive-based approaches for Mode I delamination modeling. While the present work is restricted to two-dimensional analyses with linear elastic material behavior and idealized interfaces, the findings provide a clear and robust reference for method selection in composite fracture simulations. Future extensions will address mixed-mode delamination, three-dimensional effects, nonlinear constituent behavior, and experimental validation, thereby enabling broader applicability of the proposed comparative framework to practical composite fracture problems.

Appendix

A Continuum Origin of the VCCT

The VCCT follows directly from the continuum definition of the mechanical work required to close a virtually extended crack. Consider an interface crack of initial length a_0 that is extended by a small virtual increment Δa . If the effect of this virtual extension on the local stress and displacement fields is assumed negligible [54], the work W needed to close the newly created crack faces may be written in integral form as

$$W = \frac{1}{2} \left[\int_0^{\Delta a} \sigma_{yy}^{(a_0)}(x) \delta u_y^{(a_0)}(x - \Delta a) dx + \int_0^{\Delta a} \sigma_{yx}^{(a_0)}(x) \delta u_x^{(a_0)}(x - \Delta a) dx + \int_0^{\Delta a} \sigma_{yz}^{(a_0)}(x) \delta u_z^{(a_0)}(x - \Delta a) dx \right], \quad (20)$$

where the superscript (a_0) denotes that all fields correspond to the original crack configuration prior to virtual extension. The stress components $\sigma_{y\alpha}$ ($\alpha = x, y, z$) represent the

tractions acting on the interface plane whose outward normal is aligned with the y -direction, and δu_α are the relative crack-face displacements evaluated behind the crack tip.

Following the classical argument of Rybicki and Kanninen [70], finite element discretization converts the integral in Eq. (20) into nodal force-displacement pairs, producing the discrete form of the closing work,

$$W = \frac{1}{2} \mathbf{F}^{cd} \cdot (\mathbf{u}^c - \mathbf{u}^d), \quad (21)$$

where $\mathbf{F}^{cd}$ is the nodal reaction force at the first node pair (c, d) ahead of the crack tip, and $\mathbf{u}^c$ and $\mathbf{u}^d$ are the corresponding upper and lower crack-tip displacements behind the tip. Equation (21) is the standard form implemented in commercial finite element solvers such as ABAQUS, ANSYS, and LS-DYNA, as well as in the present work.

The associated continuum strain-energy release rate follows as

$$G = \lim_{\Delta a \rightarrow 0} \frac{W}{\Delta A} = \lim_{\Delta a \rightarrow 0} \frac{1}{2b \Delta a} \mathbf{F}^{cd} \cdot (\mathbf{u}^c - \mathbf{u}^d), \quad (22)$$

where $\Delta A = b \Delta a$ is the newly created fracture surface area. In the two-dimensional DCB configuration considered in this study, only the opening mode contributes, and Eq. (22) reduces directly to the Mode I expression reported in the main text. This appendix is included to make explicit the continuum foundations of the nodal VCCT formulation used throughout the numerical simulations.

B Derivation of the Mode I Analytical Solution

This appendix provides the complete derivation of the analytical expressions summarized in Section 2.2.

B.1 Cantilever beam deflection

Each arm of the DCB is treated as a cantilever beam of physical crack length a , with effective length $\bar{a} = a + \Delta_c$. The bending moment distribution is

$$M(x) = P(\bar{a} - x), \quad (23)$$

and the strain energy stored in one arm is

$$U_A = \int_0^{\bar{a}} \frac{M^2(x)}{2E_1 I} dx = \frac{P^2 \bar{a}^3}{6E_1 I}, \quad (24)$$

where the second moment of area is $I = Bh^3/12$. Substituting I gives

$$U_A = \frac{2P^2 \bar{a}^3}{E_1 Bh^3}. \quad (25)$$

Because the DCB specimen contains two identical arms, the total strain energy is

$$U_{\text{DCB}} = 2U_A = \frac{4P^2 \bar{a}^3}{E_1 Bh^3}. \quad (26)$$

The full load–displacement relation follows from differentiation:

$$\delta = \frac{\partial U_{\text{DCB}}}{\partial P} = \frac{4P\bar{a}^3}{E_1 B h^3}. \quad (27)$$

Because the DCB specimen contains two identical arms loaded symmetrically, all analytical force–displacement relations used in the main text correspond to the full DCB, obtained by doubling the single-arm strain energy before differentiation.

B.2 Mode I energy release rate

The global Mode I energy release rate is defined as

$$G_I = \frac{1}{B} \frac{dU_{\text{DCB}}}{da}. \quad (28)$$

With $\bar{a} = a + \Delta_c$ and differentiating (26), one obtains

$$G_I = \frac{12P^2\bar{a}^2}{E_1 B^2 h^3}. \quad (29)$$

B.3 Critical (peak) load and displacement

Applying the Griffith condition $G_I = G_{Ic}$ yields the critical (peak) load

$$P_c = \sqrt{\frac{G_{Ic} E_1 B^2 h^3}{12\bar{a}^2}}. \quad (30)$$

The corresponding displacement is found by substituting P_c into Eq. (27):

$$\delta_c = \frac{4P_c\bar{a}^3}{E_1 B h^3}. \quad (31)$$

B.4 Crack-tip rotation correction

To account for the lack of perfect rotational restraint at the crack tip in orthotropic laminates, the crack-tip rotation correction of Williams [81] and Pereira & de Moraes [65] is used:

$$\Delta_c = h \sqrt{\frac{E_1}{11G_{12}} \left[3 - 2 \left(\frac{\Gamma}{1 + \Gamma} \right)^2 \right]}, \quad (32)$$

with

$$\Gamma = 1.18 \sqrt{\frac{E_1 E_2}{G_{12}}}. \quad (33)$$

The effective crack length $\bar{a} = a + \Delta_c$ is used in all final expressions for the analytical load–displacement response.

C Plane Stress versus Plane Strain Verification

To assess the influence of the two-dimensional modeling assumptions, a comparison between plane-stress and plane-strain formulations has been performed for representative DCB simulations. Plane-stress elements (CPS4I) and plane-strain elements (CPE4I) are used with identical geometry, material properties, cohesive parameters, and loading conditions. The resulting load-displacement responses are compared in Fig. 11.

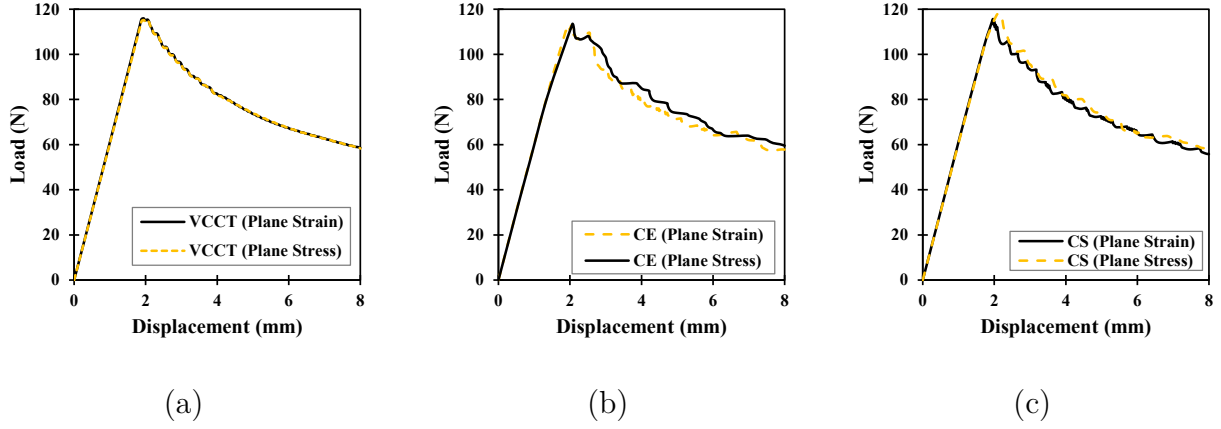


Figure 11: Comparison of global load-displacement responses obtained using plane-stress (CPS4I) and plane-strain (CPE4I) formulations for representative DCB simulations. The predicted peak loads differ by less than 2%, indicating negligible sensitivity of the global response to the choice of two-dimensional assumption for the present geometry.

Abbreviations, symbols and acronyms

a_0	Initial crack length
a	Current (dynamic) crack length
$\bar{a}$	Effective crack length
b	Width of the crack (Section 2.1.1)
B	Width of the beam (Section 2.2)
CZM	Cohesive zone method
DCB	Double cantilever beam
Δ	Separation vector
Δ_n	Normal separation component (Mode I)
Δ_s	Shear separation component (Mode II)

Δ_t	Tearing separation component (Mode III)
ΔA	Increment of crack surface area
Δa	Increment of crack length
Δ_c	Crack-tip correction displacement
Δ_{n1}	Separation at damage initiation
Δ_{nc}	Separation at complete failure
δ	Mid-span deflection of the DCB
δ_c	Critical deflection of the DCB
E_1	Longitudinal elastic modulus
E_2	Transverse in-plane elastic modulus
η	BK material parameter
$\mathbf{F}^{cd}$	Crack-closure force
G_{12}	In-plane shear modulus
G	Strain energy release rate
G_c	Critical fracture energy (fracture toughness)
G_{Ic}	Critical Mode I fracture energy
G_{IIc}	Critical Mode II fracture energy
G_{IIIc}	Critical Mode III fracture energy
G_I	Mode I strain energy release rate
G_{II}	Mode II strain energy release rate
G_{III}	Mode III strain energy release rate
h	Thickness of one arm of the DCB
P	Applied load
P_c	Critical load
Γ	Crack-tip correction parameter
$\mathbf{t}$	Nominal traction vector
t_n	Normal traction component (Mode I)

t_n^0	Peak normal traction
t_s	Shear traction component (Mode II)
t_s^0	Peak shear traction
t_t	Tearing traction component (Mode III)
t_t^0	Peak tearing traction
$\mathbf{u}^c$	Displacement vector of the upper node
$\mathbf{u}^d$	Displacement vector of the lower node
U_{DCB}	Strain energy of the DCB specimen
VCCT	Virtual crack closure technique
W	Virtual work

Data availability

All input files used in this study are available at https://github.com/asurovi/Composite_Delmination_Undergrad_Article1.

Funding

No funding was received for this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions

Agnila Ghosh Surovi: Conceptualization, Methodology, Investigation, Writing – original draft. **Md. Abdul Hasib:** Supervision, Conceptualization, Writing – review & editing. **Anik Das Anto:** Investigation, Data curation, Visualization, Writing – review & editing. **Arup Kumar Debnath:** Conceptualization, Methodology, Writing – review & editing. **Md. Ashraful Islam:** Formal analysis, Data curation, Writing – review & editing.

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